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IMAGE SENSOR AND MANUFACTURING METHOD THEREOF

Abstract

A method includes forming a doped region in a substrate; forming a doped well vertically extending from a front side of the substrate into the doped region; forming a transfer gate over the front side of the substrate and laterally between a first portion of the doped well and a second portion of the doped well; forming a trench vertically extending from the front side of the substrate through the doped well toward a back side of the substrate; depositing a dielectric material over the front side of the substrate and in the trench; recessing the dielectric material; performing a selective epitaxial growth process to form a sealing layer in the trench and over the recessed dielectric material from the front side of the substrate; forming an electrode in the trench and over the sealing layer from the front side of the substrate.

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Background/Summary

BACKGROUND

[0001] Semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs and, for these advances to be realized, similar developments in IC processing and manufacturing are needed.

[0002] In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling-down also produces a relatively high power dissipation value, which may be addressed by using low power dissipation devices such as complementary metal-oxide-semiconductor (CMOS) devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1A illustrates a top schematic view of an image sensor in accordance with some embodiments of the present disclosure.

[0005] FIG. 1B illustrates a cross-sectional view taken along line A-A' as shown in FIG. 1A.

[0006] FIG. 1C is a local enlarged view of a region A0 in FIG. 1B.

[0007] FIG. 1D illustrates a block diagram of an integrated chip (IC) including the image sensor of FIGS. 1A-1C in accordance with some embodiments of the present disclosure.

[0008] FIGS. 2-10A, 11A, 12A, 13-15A, 16-25A, and 26-28A illustrate the cross-sectional views of intermediate stages in the formation of an image sensor in accordance with some embodiments.

[0009] FIGS. 10B, 11B, 12B, 15B, 25B, and 28B are local enlarged views of regions A1, A2, A3, A4, A5, and A6 in FIGS. 10A, 11A, 12A, 15A, 25A, and 28A, respectively.

[0010] FIGS. 12C and 25C illustrate local enlarged cross-sectional views taken along line B-B' as shown in FIG. 1A and corresponding to FIGS. 12B and 25B in accordance with some embodiments of the present disclosure.

[0011] FIG. 29 is a cross-sectional view of a plasma processing apparatus in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course,

merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures.

[0014] The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated **90** degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. As used herein, “around,” “about,” “approximately,” or “substantially” may generally mean within **20** percent, or within **10** percent, or within **5** percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around,” “about,” “approximately,” or “substantially” can be inferred if not expressly stated. One skilled in the art will realize, however, that the values or ranges recited throughout the description are merely examples, and may be reduced or varied with the down-scaling of the integrated circuits.

[0015] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0016] The evolution of CMOS Image Sensors (CIS) has led to the advent of the small-pixel floating diffusion trench isolation (FDTI)-CIS. One of the objectives in the development of small-pixel FDTI-CIS is achieving pixel shrinkage while maintaining high image quality and sensor performance, and thus the floating diffusion (FD) region can be positioned at the crossroads of the pixel architecture. Polysilicon (poly-Si) has been employed as the sealing layer in FDTI-CIS configuration. However, this approach has shown limitations in the small-pixel FDTI-CIS (e.g., the infiltration of poly-Si into the seams of the FDTI's oxide layer), which in turn adversely affects the image sensor's performance, leading to potential defects and inconsistency in the sensor output.

[0017] Therefore, the present disclosure in various embodiments provides a FDTI-CIS using selective silicon epitaxial growth (SEG) for the sealing layer to prevent polysilicon deposition in the oxide seam of FDTI structure. The SEG can be utilized for control of the sealing layer's profile over the FDTI structure, ensuring accurate layer formation. The sealing layer can be in a crystalline state to have a higher etching resistance to prevent the etchant from damaging the dielectric layer over the sealing layer. In addition, before forming the sealing layer over the oxide of the FDTI structure, a dry etch process (e.g., capacitively coupled plasma) can create a thin oxide layer on the FDTI sidewall. Subsequently, a pre-clean process (e.g., SiCoNi) followed by cyclic deposition and annealing can be incorporated to scale down the thin oxide layer. Therefore, the sealing layer can form a flat Si-EPI merged-plane over the thin oxide layer, which in turn allowing for yielding a high-quality sealing layer, and thus the performance of the image sensor can be enhanced.

[0018] Reference is made to FIGS. **1A-1C**. FIG. **1A** illustrates a top schematic view of an image sensor in accordance with some embodiments of the present disclosure. FIG. **1B** illustrates a cross-sectional view taken along line A-A' as shown in FIG. **1A**. FIG. **1C** is a local enlarged view of a region **A0** in FIG. **1B**.

[0019] The image sensor **100** can include a substrate **102**. The substrate **102** has a front side **102f** and a back side **102b** opposite the front-side **102f**. In some embodiments, the front side **102f** of the substrate **102** is defined by a first surface (e.g., a front-side surface), and the back side **102b** of the substrate **102** is defined by a second surface (e.g., a back-side surface) that is opposite the first surface. The substrate **102** comprises a pixel region **103**. The pixel region **103** is a portion of the substrate **102** in which features (e.g., structural features that are described in more detail below) of an individual pixel (e.g., pixel sensor) of the image sensor are disposed. In some embodiments, the substrate **102** can include a plurality of pixel regions **103** that are disposed in an array, and the pixel region **103** is an individual pixel region of the plurality of pixel regions **103**. The substrate **102** may have a first doping type (e.g., p-type/n-type), or may be intrinsic. A pad layer **101** can be formed over the substrate **102**. By way of example and not limitation, forming of the pad layer **101** can be performed prior to forming the first and second doped regions **104** and **106**. In some embodiments, the pad layer **101** can be an oxide layer, such as silicon oxide (SiO₂).

[0020] A first doped region **104** is formed in the substrate **102**. The first doped region **104** is also disposed in the pixel region **103**. The first doped region **104** is a region of the substrate **102** having a second doping type (e.g., n-type/p-type). The second doping type is opposite the first doping type (e.g., the second doping type is n-type and the first doping type is p-type, or vice versa). In some embodiments, the first doped region **104** is implanted with second doping type dopants (e.g., n-type dopants (such as phosphorus (P), arsenic (As), antimony (Sb), etc.) or p-type dopants (such as boron (B), aluminum (Al), gallium (Ga), etc.)).

[0021] A second doped region **106** is formed in a substrate **102**. The second doped region **106** is formed over the first doped region **104**. The second doped region **106** is also formed in the plurality of pixel regions **103**. The second doped region **106** is a region of the substrate **102** having the second doping type. In some embodiments, the first doped region **104** has a greater concentration of the second doping type dopants than the second doped region **106**. In some embodiments, the first doped region **104** has a lower concentration of the second doping type dopants than the second doped region **106**.

[0022] A doped well **124** is formed in the substrate **102**. The doped well **124** can be a portion of the substrate **102** having the first doping type (e.g., p-type dopants). The doped well **124** can be formed extending vertically into the substrate **102**. The doped well **124** can be formed extending vertically into the second doped region **106**. The doped well **124** can be formed extending laterally through the substrate **102**, such that the doped well laterally surrounds the plurality of pixel regions **103**. In some embodiments, the doped well **124** can be formed extending partially through the substrate **102**. In some embodiments, the doped well **124** can be formed extending vertically into the second doped region **106** from the front side **102f** of the substrate **102** and formed vertically spaced from the back side **102b** of the substrate **102**.

[0023] A third doped region **108** can be formed in the pixel region **103**. The third doped regions **108** can be formed over the second doped region **106**. The third doped region **108** can be the region of the substrate **102** having the second doping type. In some embodiments, the third doped regions may have a greater concentration of the second doping type dopants than the second doped region **106**. In some embodiments, the third doped region **108** can have a lower concentration of the second doping type dopants than the second doped region **106**. A fourth doped regions **110** can be formed in the pixel region **103**. The fourth doped region **110** can be formed over the third doped region **108**. The fourth doped region **110** can be the region of the substrate **102** having the first doping type.

[0024] Nodes **112** can be formed in the doped well **124**. The nodes **112** can be a part of the pixel architecture in the image sensor **100**. The node **112** having the second doping type can act as a charge-to-voltage conversion site, where the electrons generated by the photodiode are transferred and converted into a voltage signal. The nodes **112** may also be disposed, at least partially, in the pixel region **103**. The nodes **112** may be laterally spaced from the third doped region **108** and/or the

fourth doped region **110**. The node **112** having the second doping type can be referred to as a common floating diffusion node, and the other node **112** at a side of the transfer gate **114** opposite to the node **112** having the first doping type can be referred to as a grounding node. In some embodiments, the node **112** can have a greater concentration of the first/second doping type dopants than the first doped region **104**, the second doped region **106**, and/or the third doped region **108**. In some embodiments, the node **112** having the second doping type may be a common floating diffusion node that is shared by two or more pixel sensor unit.

[0025] A transfer gate **114** is over/on the front side **102f** of the substrate **102**. In some embodiments, the transfer gate **104** can be formed between two portions of the doped well **124**. The transfer gate **114** can include a gate dielectric structure **116** and a gate electrode structure **118**. In some embodiments, the gate dielectric structure **116** can be made of, for example, an oxide (e.g., silicon dioxide (SiO₂)), a high-k dielectric material (e.g., hafnium oxide (HfO), tantalum oxide (TaO), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), aluminum oxide (AlO), zirconium oxide (ZrO), some other dielectric material with a dielectric constant greater than about 3.9), some other dielectric material, or a combination of the foregoing. The gate electrode structure **118** can be made of, for example, polysilicon, a metal (e.g., aluminum (Al), copper (Cu), titanium (Ti), tantalum (Ta), tungsten (W), molybdenum (Mo), cobalt (Co), or the like), some other conductive material, or a combination of the foregoing.

[0026] A semiconductor layer **147** is formed in the node **112** and near to the front side **102f** of the substrate **102**. The semiconductor layer **147** can serve as an electrode of the node **112**. The semiconductor layer **147** can be made of polycrystalline silicon and have ability to conformally coat irregular surfaces (e.g., the inside of the trench **102t**), and its electrical properties, which can be suitable for sensor applications. In some embodiments, the semiconductor layer **147** can have a top surface higher than the front-side surface of the pad layer **101** and the front-side surface of the substrate **102**. The sealing layer **127** can have a footprint that is disposed within a footprint of a dielectric layer **137** and/or a footprint of the semiconductor layer **147**.

[0027] The sealing layer **127** is formed in the substrate **102** and between the semiconductor layer **147** and the deep trench isolation structure **150**. The sealing layer **127** can be used as an etch stop layer when etching the dielectric structure **125** (see FIG. 23) from the back side **102b** of the substrate **102** to prevent the etchant from damaging the dielectric layer **137** (see FIG. 23). In some embodiments, the sealing layer **127** is to be crystallized (e.g., by using a thermal crystallization process), and thus the sealing layer **127** can be in a crystalline state to have a higher etching resistance. By way of example and not limitation, the sealing layer **127** can be made of monocrystalline silicon, which is a single, continuous crystal lattice. As shown in FIGS. 1C, the sealing layer **127** can have a flat bottom surface **127b**. The flat bottom surface **127** of the sealing layer **127** can enhance its ability to act as an effective barrier against etchants during the etching of the dielectric structure **125** (see FIG. 23) from the back side **102b** of the substrate **102**. In some embodiments, the sealing layer **127** may have a top surface level with a front-side surface of the pad layer **101** or a front-side surface of the substrate **102**. In some embodiments, the sealing layer **127** may be a grid structure.

[0028] A dielectric layer **137** is sandwiched between the sealing layer **127** and the semiconductor layer **147**. The dielectric layer **137** can act as an insulator, preventing electrical crosstalk and leakage between the semiconductor layer **147** and the sealing layer **127**. The dielectric layer **137** can have a top surface lower than the front-side surface of the substrate **102**. In some embodiments, the dielectric layer **137** can have a top surface in a position between a lower boundary of the node **112** and an upper boundary of the node **112**. In some embodiments, the dielectric layer **137** can have a level back-side surface conformal to the front-side surface of the sealing layer **127**. In some embodiments, the dielectric layer **137** can have a level top surface in parallel with the level bottom surface thereof. By way of example and not limitation, the dielectric layer **137** may be made of a dielectric material such as silicon nitride (SiN), silicon oxide (SiO₂), silicon carbo-nitride

(SiCN), silicon oxynitride (SiON), silicon oxy-carbo-nitride (SiOCN), or the like. In some embodiments, the dielectric layer **137** may be a grid structure.

[0029] An ILD structure **140** is formed over the front side **102f** of the substrate **102** and over the transfer gate **114**. An interconnect structure **142** is formed in the ILD structure **140** and over the front side **102f** of the substrate **102**. In some embodiments, the interconnect structure **142** can include a plurality of conductive contacts **142a**, a plurality of conductive vias **142b**, and a plurality of conductive wires **142c**.

[0030] A deep trench isolation (DTI) structure **150** is in the substrate **102** and can be used to provide electrical insulation and/or structural integrity to the image sensor **100**. The deep trench isolation structure **150** can be a grid structure when viewed from the top and include the dielectric filler **154** and the dielectric liner **152** wrapping around the dielectric filler **154**. In the cross-sectional view, the deep trench isolation structure **150** can be characterized by the positioning of the node **112**/doped well **124** directly above the deep trench isolation structure **150**. The deep trench isolation structure **150** can be a deeply etched trench structure in the substrate **102** to isolate the components of the sensor from each other, thereby reducing electronic noise and crosstalk between pixels. When viewed from the top, the deep trench isolation structure **150** can reveal that the node **112** is located at the intersection (or cross-road) of the isolation grid **156**.

[0031] The deep trench isolation structure **150** can vertical sidewalls, which indicate the sidewalls of the deep trench isolation structure **150** formed in the substrate **102** can be perpendicular to the front-side surface of the substrate **102** and/or the back-side surface of the substrate **102**. This vertical alignment can be used to create well-defined, isolated regions for each pixel in the sensor array. The deep trench isolation structure **150** can have a flat front-side surface perpendicular to the sidewalls thereof, and thus the deep trench isolation structure **150** can have a square front-side profile. This square profile is in direct contact with the sealing layer **127**. In some embodiments, the sidewall of the deep trench isolation structure **150** is coterminous to the sidewall of the sealing layer **127**.

[0032] In some embodiments, the deep trench isolation structure **150** can be interchangeably referred to a floating diffusion trench isolation (FDTI), and thus the image sensor **100** can be interchangeably referred to a FDTI-CMOS image sensor (FDTI-CIS). The deep trench isolation structure **150** is formed with a footprint that is disposed within the footprint of the doped well **124**, which in turn reduces the size of the pixel region **103**. As shown in FIGS. 25B and 25C, the sealing layer **127** is positioned between the node **112** and the deep trench isolation structure **150**.

[0033] An isolation grid **156** is formed over the back-side **102b** of the substrate **102**. In further embodiments, the isolation grid **156** can be formed overlying the deep trench isolation structure **150**. The isolation grid **156** may be or include, for example, a metal (e.g., tungsten (W), aluminum (Al), cobalt (Co), copper (Cu), silver (Ag), gold (Au), some other metal, or a combination of the foregoing), an oxide (e.g., SiO₂), a nitride (e.g., SiN), a carbide (e.g., SiC), a high-k dielectric material (e.g., HfO, TaO, etc.), a low-k dielectric material, some other isolation material, or a combination of the foregoing. A dielectric layer **158** is formed over the over the back-side surface of the dielectric filler **154** and over the isolation grid **156**. In other embodiments, the dielectric layer **158** may be formed along the back side **102b** of the substrate **102** and over the isolation grid **156**.

[0034] A plurality of electromagnetic radiation (EMR) filters **160** are formed on/over the dielectric layer **158** and within the isolation grid **156**. In some embodiments, the EMR filters **160** may include one or more light filtering materials that allow for the transmission of radiation (e.g., light) having a specific wavelength range, while blocking light of wavelengths outside of the specified range. A plurality of micro-lenses **162** are formed on/over the plurality of EMR filters **160**. The micro-lenses **162** may include a micro-lens material. The micro-lens **162** can have a curved upper surface. In some embodiments, the micro-lens **162** may comprise a photoresist material exposed using a distributing exposing light dose (e.g., for a negative photoresist more light is exposed at a bottom of the curvature and less light is exposed at a top of the curvature), developed, and baked to

form a rounding shape. In some embodiments, after the plurality of micro-lenses **162** are formed, formation of a plurality of pixel sensor units **107** (see FIG. **1D**) is complete. In further embodiments, after the plurality of micro-lenses **162** are formed, formation of the image sensor **100** can be complete. In some embodiments, the image sensor **100** can be interchangeably referred to as a back-side illumination configuration.

[0035] Reference is made to FIG. **1D**. FIG. **1D** illustrates a block diagram of an integrated chip (IC) **200** including the image sensor **100** of FIGS. **1A-1C** in accordance with some embodiments of the present disclosure. The IC **200** can include a first chip **201**, a second chip **202**, and a third chip **203**. In some embodiments, the first chip **201** comprises the image sensor of the present disclosure. For example, in some embodiments, the first chip **201** can include the plurality of pixel sensor units **170**. The first chip **201**, the second chip **202**, and the third chip **203** can be bonded together (e.g., via one or more bonding structures). The first chip **201**, the second chip **202**, and the third chip **203** can be vertically stacked and electrically coupled together (e.g., via one or more conductive pads). In such embodiments, the image sensor **100** may be referred to as a three chip image sensor (e.g., 3-chip CIS). While the block diagram of FIG. **1D** illustrates the IC **200** comprising three chips that are bonded together, it will be appreciated that the IC **200** may comprise any number of chips bonded together (e.g., 2 chips, 3 chips, 4 chips, 5 chips, etc.). It will also be appreciated that, in some embodiments, the IC **200** may only include the first chip **201** (e.g., 1-chip CIS).

[0036] Reference is made to FIGS. **2-28B**. FIGS. **2-10A**, **11A**, **12A**, **13-15A**, **16-25A**, and **26-28A** illustrate the cross-sectional views of intermediate stages in the formation of an image sensor **100** in accordance with some embodiments. FIGS. **10B**, **11B**, **12B**, **15B**, and **25B** are local enlarged views of regions **A1**, **A2**, **A3**, **A4**, **A5**, and **A6** in FIGS. **10A**, **11A**, **12A**, **15A**, **25A**, and **28A**, respectively. It is understood that additional operations can be provided before, during, and after the processes shown by FIGS. **2-28B**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable.

[0037] Reference is made to FIG. **2**. The image sensor **100** can include a substrate **102**. The substrate **102** has a front side **102f** and a back side **102b** opposite the front-side **102f**. In some embodiments, the front side **102f** of the substrate **102** is defined by a first surface (e.g., a front-side surface), and the back side **102b** of the substrate **102** is defined by a second surface (e.g., a back-side surface) that is opposite the first surface. The substrate **102** comprises a pixel region **103**. The pixel region **103** is a portion of the substrate **102** in which features (e.g., structural features that are described in more detail below) of an individual pixel (e.g., pixel sensor) of the image sensor are disposed. In some embodiments, the substrate **102** can include a plurality of pixel regions **103** that are disposed in an array, and the pixel region **103** is an individual pixel region of the plurality of pixel regions **103**.

[0038] The substrate **102** may include any type of semiconductor body (e.g., monocrystalline silicon/complementary metal-oxide-semiconductor (CMOS) bulk, germanium (Ge), a group III-V semiconductor material, silicon-germanium (SiGe), silicon on insulator (SOI), etc.). In some embodiments, the image sensor (e.g., back-side illumination image sensor) **100** can be configured to record incident radiation (e.g., photons) that passes through the back side **102b** of the substrate **102**. In some embodiments, the image sensor (e.g., front-side illumination image sensor) is configured to record incident radiation (e.g., photons) that passes through the front side **102f** of the substrate **102**. The substrate **102** may have a first doping type (e.g., p-type/n-type), or may be intrinsic. A pad layer **101** can be formed over the substrate **102**. By way of example and not limitation, forming of the pad layer **101** can be performed prior to forming the first and second doped regions **104** and **106**. In some embodiments, the pad layer **101** can be an oxide layer, such as silicon oxide (SiO₂).

[0039] A first doped region **104** is formed in the substrate **102**. The first doped region **104** is also disposed in the pixel region **103**. The first doped region **104** is a region of the substrate **102** having

a second doping type (e.g., n-type/p-type). The second doping type is opposite the first doping type (e.g., the second doping type is n-type and the first doping type is p-type, or vice versa). In some embodiments, the first doped region **104** is formed via a blanket doping process that implants second doping type dopants (e.g., n-type dopants (such as phosphorus (P), arsenic (As), antimony (Sb), etc.) or p-type dopants (such as boron (B), aluminum (Al), gallium (Ga), etc.)) into the substrate **102**. The blanket doping process does not utilize a masking layer (e.g., positive/negative photoresists, hardmask, etc.) to implant the second doping type dopants into the plurality of pixel regions **103** of the substrate **102**. In some embodiments, the blanket doping process can be or include, for example, a blanket ion implantation process, a blanket diffusion process, some other blanket doping process, or a combination of the foregoing. It will be appreciated that a blanket doping process is different than a doping process that utilizes a masking layer to selectively implant the second doping type dopants into the plurality of pixel regions **103** of the substrate **102**. It will also be appreciated that, in other embodiments, the first doped region **104** may be formed by a doping process that utilizes a masking layer to selectively implant the second doping type dopants into the plurality of pixel regions **103** of the substrate **102**.

[0040] A second doped region **106** is formed in a substrate **102**. The second doped region **106** is formed over the first doped region **104**. The second doped region **106** is also formed in the plurality of pixel regions **103**. The second doped region **106** is a region of the substrate **102** having the second doping type. In some embodiments, the first doped region **104** has a greater concentration of the second doping type dopants than the second doped region **106**. In some embodiments, the first doped region **104** has a lower concentration of the second doping type dopants than the second doped region **106**. In some embodiments, the second doped region **106** is formed via a blanket doping process that implants second doping type dopants into the substrate **102**. The blanket doping process does not utilize a masking layer (e.g., positive/negative photoresists, hardmask, etc.) to implant the second doping type dopant into the plurality of pixel regions **103** of the substrate **102**. In some embodiments, the blanket doping process can be or include, for example, a blanket ion implantation process, a blanket diffusion process, some other blanket doping process, or a combination of the foregoing. It will be appreciated that a blanket doping process is different than a doping process that utilizes a masking layer to selectively implant the second doping type dopants into the plurality of pixel regions **103** of the substrate **102**. It will also be appreciated that, in other embodiments, the second doped region **106** may be formed by a doping process that utilizes a masking layer to selectively implant the second doping type dopants into the plurality of pixel regions **103** of the substrate **102**.

[0041] Reference is made to FIG. 3. A doped well **124** is formed in the substrate **102**. The doped well **124** can be a portion of the substrate **102** having the first doping type. The doped well **124** can be formed extending vertically into the substrate **102**. The doped well **124** can be formed extending vertically into the second doped region **106**. The doped well **124** can be formed extending laterally through the substrate **102**, such that the doped well laterally surrounds the plurality of pixel regions **103**. In some embodiments, the doped well **124** can be formed extending partially through the substrate **102**. In some embodiments, the doped well **124** can be formed extending vertically into the second doped region **106** from the front side **102f** of the substrate **102** and formed vertically spaced from the back side **102b** of the substrate **102**.

[0042] In some embodiments, the doped well **124** may be formed by a doping process (e.g., via ion implantation, diffusion, etc.) that utilizes a patterned masking layer (not shown) (e.g., positive/negative photoresist, a hardmask, etc.) on the front side **102f** of the substrate **102** to selectively implant first doping type dopants (e.g., p-type dopants) into the substrate **102**. In some embodiments, a process for forming the patterned masking layer can include depositing a masking layer (not shown) on the front side **102f** of the substrate **102**. The masking layer may be deposited by, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), a spin-on process, some other deposition process, or a combination of the

foregoing. Thereafter, the masking layer is exposed to a pattern (e.g., via a lithography process, such as photolithography, extreme ultraviolet lithography, or the like) and developed, thereby forming the patterned masking layer over the front side **102f** of the substrate **102**. With the patterned masking layer in place over the front side **102f** of the substrate **102**, the doping process is performed on the substrate **102**, thereby forming the doped well **124**.

[0043] In some embodiments, the doping process may be, for example, an ion implantation process, a diffusion process, some other doping process, or a combination of the foregoing. Subsequently, in some embodiments, the patterned masking layer **1402** is stripped away. Because the second doped region **106** can be formed via a blanket doping process, rather than a doping process that utilized a lithography process, the doped well **124** may be formed in contact with the second doped region **106**.

[0044] A third doped region **108** can be formed in the pixel region **103**. The third doped regions **108** can be formed over the second doped region **106**. The third doped region **108** can be the region of the substrate **102** having the second doping type. In some embodiments, the third doped regions may have a greater concentration of the second doping type dopants than the second doped region **106**. In some embodiments, the third doped region **108** can have a lower concentration of the second doping type dopants than the second doped region **106**. In some embodiments, a process for forming the plurality of third doped region **108** can include forming a patterned masking layer (not shown) (e.g., negative/positive photoresist, a hardmask, etc.) over the front side **102f** of the substrate **102**. Thereafter, with the patterned masking layer in place, a doping process (e.g., an ion implantation process, a diffusion process, etc.) is performed on the substrate to selectively implant second doping type dopants into the substrate **102** according to the patterned masking layer, thereby forming the plurality of third doped regions. Subsequently, in some embodiments, the patterned masking layer is stripped away.

[0045] A fourth doped regions **110** can be formed in the pixel region **103**. The fourth doped region **110** can be formed over the third doped region **108**. The fourth doped region **110** can be the region of the substrate **102** having the first doping type. In some embodiments, the plurality of fourth doped regions may be formed by a doping process (e.g., via (angled) ion implantation, diffusion, etc.) that utilizes a patterned masking layer (not shown) (e.g., positive/negative photoresist, a hardmask, etc.) over the front side **102f** of the substrate **102** (and over the plurality of transfer gates) to selectively implant first doping type dopants into the substrate **102**.

[0046] Nodes **112** can be formed in the doped well **124**. The nodes **112** can be a part of the pixel architecture in the image sensor **100**. The node **112** having the second doping type can act as a charge-to-voltage conversion site, where the electrons generated by the photodiode are transferred and converted into a voltage signal. The node **112** may also be disposed, at least partially, in the pixel region **103**. The node **112** may be laterally spaced from the third doped region **108** and/or the fourth doped region **110**. The node **112** having the second doping type can be referred to as a floating diffusion node, and the other node **112** at a side of the transfer gate **114** opposite to the node **112** having the first doping type can be referred to as a grounding node. In some embodiments, the node **112** has a greater concentration of the first/second doping type dopants than the first doped region **104**, the second doped region **106**, and/or the third doped region **108**. In some embodiments, the node **112** having the second doping type may be a common floating diffusion node that is shared by two or more pixel sensor unit. In some embodiments, the node **112** may be formed by a doping process (e.g., via ion implantation, diffusion, etc.) that utilizes a patterned masking layer (not shown) (e.g., positive/negative photoresist, a hardmask, etc.) over the front side **102f** of the substrate **102** (and over the plurality of transfer gates) to selectively implant second doping type dopants into the substrate **102**. In some embodiments, the plurality of transfer gates are utilized, at least partially, as the patterned masking layer. Subsequently, in some embodiments, the patterned masking layer is stripped away.

[0047] Reference is made to FIG. **4**. A process for forming the deep trench isolation (DTI) structure

150 (see FIGS. **25A-25C**) may include forming a patterned mask layer **113** (e.g., negative/positive photoresist, a hardmask, etc.) with openings **113a** over the front side **102f** of the substrate **102**. The opening **113a** may be formed with a footprint that is disposed within a footprint of the node **112**. In other words, the node **112** can be exposed from the opening **113a**. In some embodiments, the mask layer **113** can be made of nitride, such as silicon nitride. In some embodiments, the mask layer **113** can have a thickness **T1**. In some embodiments, the opening **113a** can have a lateral dimension **D1**. [0048] Reference is made to FIG. **5**. With the patterned mask layer **113** in place over the front side **102f** of the substrate **102**, an etching process **P1** is then performed on the substrate **102**. The etching process **P1** can remove unmasked portions of the substrate **102**, thereby forming a trench **102t** in the substrate **102**. The trench **102t** extends partially through the substrate **102** (e.g., not fully through the substrate **102**) from the front side **102f** of the substrate **102**, thereby extending vertically through doped well **124** and the second doped region **106** and vertically into the first doped region **104**. In such embodiments, the trench **102t** can expose portions of the first doped region **104** and portions of the second doped region **106**. In further such embodiments, the portions of the first doped region **104** and the portions of the second doped region **106** at least partially define surfaces (e.g., sidewalls) of the trench **102t**. In some embodiments, the trench **102t** may have angled sidewalls. In other embodiments, the sidewalls of the trench **102t** may be substantially straight (e.g., vertical).

[0049] The trench **102t** is formed extending laterally through the substrate **102**, such that the trench **102t** laterally surrounds the plurality of pixel regions **103**. The trench **102t** is formed with a footprint that is disposed within the footprint of the doped well **124**. In some embodiments, the second doped region **106** continuously extends laterally between opposite inner sidewalls of the trench **102t**. In some embodiments, the second doped region **106** continuously extends laterally between the opposite inner sides of the doped well **124** and continuously extends laterally between the opposite inner sidewalls of the trench **102t**. In some embodiments, the first doped region **104** continuously extends laterally between the opposite inner sidewalls of the trench **102t**. In some embodiments, a layout of the trench **102t** has a grid-like shape. As such, the footprint of the trench **102t** has the grid-like shape. In such embodiments, the grid-like shaped footprint of the trench **102t** is disposed within a grid-like shaped footprint of the doped well **124**.

[0050] The etching process **P1** may be or comprise, for example, a wet etching process, a dry etching process, a reactive ion etching (RIE) process, some other etching process, or a combination of the foregoing. In some embodiments, the trench **102t** can have a lateral dimension (or width) **D2** and a vertical dimension (or depth) **D3**, and a ratio of the vertical dimension **D3** to the lateral dimension **D2** can be in a range from about 10 to 50, such as about 10, 15, 20, 25, 30, 35, 40, 45, or 50. By way of example and not limitation, the lateral dimension **D2** can be in a range from about 1.5 to 6 nm, such as about 1, 5, 2, 2.5, 3, 3.2, 3.5, 4, 4.5, 5, 5.5, or 6 nm. The vertical dimension **D3** can be in a range from about 50 to 150 nm, such as about 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, or 150 nm. In some embodiments, the etching process would consume the mask layer **113**, and thus after the etching process **P1** is complete, the mask layer **113** may have a thickness **T2** thinner than the thickness **T1** of the mask layer **113** as shown in FIG. **4**.

[0051] Reference is made to FIG. **6**. The openings **113a** in the mask layer **113** can be laterally expanded through an etching process **P2**, such that a lateral dimension **D4** (see FIG. **6**) of the opening **113a** can be enlarged to be greater than the lateral dimension **D2** of the trench **102t** as shown in FIG. **5** and the lateral dimension **D1** of the opening **113a** (see FIG. **4**). In some embodiments, the etching process **P2** would consume the mask layer **113**, and thus after the etching process **P2** is complete, the mask layer **113** may have a thickness **T3** thinner than the thickness **T2** of the mask layer **113** as shown in FIG. **5**.

[0052] Reference is made to FIG. **7**. An upper portion of the trench **102t** in the substrate **102** can be laterally expanded through an etching process **P3**, such that the upper portion of the trench **102t** has a sidewall **102s** that tapers away from the front side **102f** of the substrate **102** and form a corner

102c with a middle portion of the trench **102t**. For example, the upper portion of the trench **102t** has a lateral dimension **D5** decreasing as a vertical distance from the front side **102f** of the substrate **102** increases. Therefore, the upper portion of the trench **102t** can be enlarged to have the lateral dimension **D5** that is greater than the lateral dimension **D2** of the trench **102t** as shown in FIG. 5. In some embodiments, the etching process **P3** would consume the mask layer **113**, and thus after the etching process **P3** is complete, the mask layer **113** may have a thickness **T4** thinner than the thickness **T3** of the mask layer **113** as shown in FIG. 6. By way of example and not limitation, the lateral dimension **D5** can be in a range from about 100 to 200 nm, such as about 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 nm. The upper portion of the trench **102t** can have a vertical dimension **D6** that is in a range from about 100 to 300 nm, such as 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, or 300 nm. In some embodiments, the lateral dimension **D5** can be less than the vertical dimension **D6**.

[0053] Reference is made to FIG. 8. A dielectric layer **121** can be deposited over the front side **102f** of the substrate **102** and fill into the trench **102t**. Subsequently, a dielectric layer **123** can be deposited over the dielectric layer **121**. In some embodiments, the dielectric layer **123** can be made of a different material than the dielectric layer **121**. The dielectric layer **121** can be formed to have a seam **121s** therein within the trench **102t**. In some embodiments, the seam **121s** is situated below the upper portion of the trench **102t**. In some embodiments, the seam **121s** can be interchangeably referred to as an air gap. By way of example and not limitation, the dielectric layer **121** may be made of a dielectric material such as silicon nitride (SiN), silicon oxide (SiO.sub.2), silicon carbo-nitride (SiCN), silicon oxynitride (SiON), silicon oxy-carbo-nitride (SiOCN), or the like. The dielectric layer **123** may be made of a dielectric material such as silicon nitride (SiN), silicon oxide (SiO.sub.2), silicon carbo-nitride (SiCN), silicon oxynitride (SiON), silicon oxy-carbo-nitride (SiOCN), or the like. In some embodiments, the trench **102t** can be filled only with the dielectric layer **121** but not with dielectric layer **123**.

[0054] Reference is made to FIG. 9. A removal process **P4**, such as chemical mechanical polishing (CMP), or the like, can be performed on the dielectric layers **121** and **123** until the mask layer **113** is exposed. In some embodiments, the removal process **P4** would consume the mask layer **113**, and thus after the removal process **P4** is complete, the mask layer **113** may have a thickness **T5** thinner than the thickness **T4** of the mask layer **113** as shown in FIG. 7. In some embodiments, remainders of the dielectric layers **121** and **123** as illustrated in FIG. 9 can be collectively referred to a multi-oxide structure, a multi-dielectric structure, a sacrificial dielectric, a sacrificial oxide, or a dielectric structure **125**.

[0055] Reference is made to FIGS. 10A and 10B. An etching process **P5** is performed on the dielectric structure **125** to recess the dielectric structure **125**, such that the upper portion of the trench **102t** can reappear and the seam **121s** can be exposed from the dielectric layer **121**. After the etching process **P5** is complete, a dielectric remainder **125r** (see FIG. 10B) can be formed on a top portion of the dielectric structure **125**. In other words, the dielectric remainder **125r** of the dielectric structure **125** can be formed on a bulk portion **125b** (see FIG. 10B) of the dielectric structure **125**. As shown in FIG. 10B, the top portion of the dielectric structure **125** has a slant surface **125s** and a level surface **125v** connected to the slant surface **125s**, and the slant surface **125s** can be coterminous with the sidewall **102s** of the trench **102**. In some embodiments, the dielectric remainder **125r** can be interchangeably referred to a sharp corner, a top portion, an oxide remainder, or a thin oxide layer, or a protruding portion. By way of example and not limitation, the dielectric remainder **125r** can have a vertical dimension **D7** and a lateral dimension **D8**, and the lateral dimension **D8** can be in a range from about 1 to 10 nm, such as about 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nm.

[0056] In some embodiments, the etching process **P5** can be an anisotropic etching process. In some embodiments, the etching process **P5** can be a dry etching process, such as a plasma etching process. The plasma etching process may be performed by a plasma processing apparatus 5, such

as an inductively coupled plasma (ICP) apparatus, a capacitively coupled plasma (CCP) apparatus, or a micro wave apparatus. In greater detail, Referring now to FIG. 29, illustrated is a cross-sectional view of an exemplary plasma processing apparatus 5 in some embodiments of the present disclosure. In some embodiments, the plasma processing apparatus 5 may contain an inductively-coupled plasma as a plasma source and a RF power supply as a bias power source. As shown in FIG. 29, the plasma processing apparatus 5 includes a chamber base 12 having a typically grounded chamber wall 14. The chamber base 12 is closed by a removable lid or a lid 42 and contains a pedestal assembly 48 which can typically be raised and lowered on a shaft 40 by actuation of a pedestal lift assembly 46. An inductively-coupled plasma coil 24 surrounds the lid 42 and is connected to an RF source power supply 26. The pedestal assembly 48 is connected, through an RF match network 30 which matches impedances, to an RF power supply 28. During operation of the plasma processing apparatus 5, the pedestal assembly 48 supports the substrate 102 in the chamber base 12. A plasma-generating source gas, such as argon, is introduced into the plasma processing apparatus 5 by a gas supply (not shown). Volatile reaction products and unreacted plasma species are removed from the plasma processing apparatus 5 by a gas removal mechanism (not shown). Source power such as a high voltage signal, provided by the RF source power supply 26, is applied to the inductively-coupled plasma coil 24 to ignite and sustain the plasma in the plasma processing apparatus 5. Ignition of the plasma in the plasma processing apparatus 5 is accomplished primarily by electrostatic coupling of the inductively-coupled plasma coil 24 with the source gas, due to the large-magnitude voltage applied to the inductively-coupled plasma coil 24 and the resulting electric fields produced in the plasma processing apparatus 5. Once ignited, the plasma is sustained by electromagnetic induction effects associated with time-varying magnetic fields produced by the alternating currents applied to the inductively-coupled plasma coil 24. Through the RF power supply 28, the pedestal assembly 48 is typically electrically biased to provide to the substrate 102 ion energies that are independent of the RF voltage applied to the chamber 10 through the inductively-coupled plasma coil 24 and RF source power supply 26. This facilitates more precise control over the energies of the etchant ions that bombard the surface of the substrate 102 to remove the unexposed portion (see FIG. 3B) of the dielectric structure 125. A bias etching can be provided by the plasma source 26 with turning on the RF power source 28 during the bias etching step.

[0057] In some embodiments, the bias power of the etching process P5 may be performed at a frequency in a range from about 13.56 MHz to about 2.45 GHz. In some embodiments, the bias power of the etching process P5 may be in a range from about 100 W to about 2000 W, such as 100, 500, 1000, 1500, or 2000 W. In some embodiments, the plasma processing apparatus 5 may also be an electron cyclotron resonance (ECR) apparatus, but the present disclosure is not limited thereto. In some embodiments, the etching process P5 may use a gas mixture including, such as C.sub.4F.sub.8, with the bias. In some embodiments, the etching process P5 may be performed under a pressure in a range from about 3 to 100 mTorr, such as about 3, 10, 20, 30, 40, 50, 60, 70, 80, 90, or 100 mTorr.

[0058] Reference is made to FIGS. 11A and 11B. An etching process P6 is performed on the dielectric structure 125 to scale down the dielectric remainder 125r of the dielectric structure 125. In some embodiments, the etching process P6 can be used to remove native oxide on the dielectric remainder 125r. In some embodiments, diluted HF, SiCoNi (including HF and NH.sub.3), or the like, may be used as the etchant in the etching process P6. In some embodiments, the etching process P6 can be an anisotropic etching process. In some embodiments, the etching process P6 can be a dry etching process, such as a plasma etching process. After the etching process P6 is complete, the dielectric remainder 125r can have a vertical dimension D9 and a lateral dimension D10, the vertical dimension D9 can be less than the vertical dimension D7 of the dielectric remainder 125r as shown in FIG. 10B, and the lateral dimension D10 can be less than the vertical dimension D8 of the dielectric remainder 125r as shown in FIG. 10B. By way of example and not

limitation, the lateral dimension D8 can be in a range from about 1 to 10 nm, such as about 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nm.

[0059] In some embodiments, the etching process P6 can act as a pre-clean process to prepare the surface for the subsequent deposition of the sealing layer 127 (see FIGS. 12A-12C), ensuring high-quality layer formation. The etching process P6 can be a gentle or less aggressive etching process to prevent excessive material removal or damage to the underlying layers. In some embodiments, the quality of epitaxial growth heavily may rely on the condition of the surface onto which it is deposited, and the etching process P6 can ensure that the surface is prepared, leading to uniform and high-quality epitaxial growth of the sealing layer 127. In some embodiments, the etching process P6 can be a pre-clean process.

[0060] Reference is made to FIGS. 12A-12C. A deposition process P7 is performed on an exposed sidewall 102s of the substrate 102 in the trench 102t to form a sealing layer 127 over the dielectric remainder 125r. The sealing layer 127 can be used as an etch stop layer when etching the dielectric structure 125 from the back side 102b of the substrate 102 (see FIG. 23) to prevent the etchant from damaging a subsequently formed dielectric layer 137 (see FIG. 23) over the front side of the sealing layer 127. In some embodiments, the sealing layer 127 is to be crystallized (e.g., by using a thermal crystallization process), and thus the sealing layer 127 can be in a crystalline state to have a higher etching resistance. By way of example and not limitation, the sealing layer 127 can be made of monocrystalline silicon, which is a single, continuous crystal lattice. In some embodiments, the deposition process P7 to form the sealing layer 127 can be the selective epitaxial growth (SEG) process. In some embodiments, the deposition process P7 can be interchangeably referred to as two-dimensional epitaxial (EPI) growth.

[0061] As shown in FIGS. 12B and 12C, the dielectric remainder 125r, positioned within the trench 102t, has a small top surface area. This minimal area can result in a correspondingly small contact area between the dielectric remainder 125r and the subsequently formed sealing layer 127. The reduced contact area can allow more precise control over the profile of the sealing layer 127, ensuring the sealing layer's effectiveness as an etch stop layer and maintaining the structural integrity of the image sensor 100. The sealing layer 127 is formed on the exposed sidewalls 102s of the substrate 102. In this location, the influence of the dielectric remainder 125r on the growth direction of the sealing layer can be minimized, leading to a more uniform and controlled layer formation. Specifically, as shown in FIG. 12B, when viewed from the cross-sectional view taken along line A-A' (e.g., xy-cut) as shown in FIG. 1A, the sealing layer 127 can grow with a merged-plane, exhibiting a (100) crystallographic facet, resulting in the sealing layer 127 with a flat bottom surface 127b. In other words, from the top view as shown in FIG. 1A, the deep trench isolation 150 structure has a grid pattern that has a grid line, and from a cross-sectional view taken along a direction intersecting with a lengthwise direction of the grid line, the sealing layer 127 has a flat bottom surface 127b. In some embodiments, the cross-sectional view of FIG. 12B passes through the intersection position of the grid lines in the grid pattern as shown in FIG. 1A. The flat bottom surface 127b of the sealing layer 127 can enhance its ability to act as an effective barrier against etchants during the etching of the dielectric structure 125 from the back side 102b of the substrate 102 (see FIG. 23). In some embodiments, the sealing layer 127 can be spaced apart from the slant surface 125s of the dielectric remainder 125r, and a void 127v can be formed between the bottom surface 127b of the sealing layer 127 and a top surface of the dielectric structure 125. In some embodiments, the void 127v can be communicated with the seam 121s in the dielectric structure 125. In some embodiments, the sealing layer 127 may have a top surface level with a front-side surface of the pad layer 101 or a front-side surface of the substrate 102. The top surface of the sealing layer 127 can be in parallel with the bottom surface 127b of the sealing layer 127.

Specifically, in FIG. 12B, when viewed from the cross-sectional view taken along line A-A' (e.g., xy-cut) as shown in FIG. 1A, the sealing layer 127 can have peripheral portions 127p and a central portion 127c between the peripheral portions 127p. A thickness 127t1 of the central portion 127c of

the sealing layer **127** can be substantially the same as the thicknesses **127t2** of the peripheral portions **127p**. In some embodiments, the sealing layer **127** may have a thickness in a range from about 20 to 100 nm, such as about 20, 30, 40, 50 60, 70, 80, 90, or 100 nm.

[0062] When the contact area between the dielectric remainder **125r** and the sealing layer **127** is large, it can significantly influence the growth direction and morphology of the sealing layer **127**. The large interface area can cause the sealing layer **127** to adopt a growth pattern that results in a merged-plane with a (111) or (311) crystallographic facet, leading to a triangular bottom surface on the sealing layer. A sealing layer with a triangular bottom surface presents challenges during the etching of the dielectric structure **125** from the back side **102b** of the substrate (see FIG. **23**). The pointed or angular nature of a triangular bottom surface makes the sealing layer more vulnerable to etchant penetration. During the etching process, the sealing layer **127** with a triangular bottom surface can be easily etched through, particularly at its thinner points. Once the etchant etch through the sealing layer, it can continue to etch into and damage the subsequent dielectric layer **137**, which is formed over the front side of the sealing layer. Damage to the dielectric layer **137** and other components due to an ineffective sealing layer can lead to, such as electrical shorts, leakage, or reduced isolation between sensor elements. To avoid these issues, the growth direction and surface profile of the sealing layer **127** can be controlled by minimizing the contact area between the sealing layer **127** and the dielectric remainder **125r**, ensuring the growth of a sealing layer with a flat bottom surface.

[0063] In some embodiments, as shown in FIG. **12C**, when viewed from the cross-sectional view taken along line B-B' (e.g., x-cut) as shown in FIG. **1A**, the back-side of the sealing layer **127** can grow to have a convex profile. In other words, from the cross-sectional view taken along the lengthwise direction of the grid line, the sealing layer **127** has a convex bottom surface **127b'**. In some embodiments, the cross-sectional view of FIG. **12C** passes through between two intersection positions of the grid lines in the grid pattern as shown in FIG. **1A**. In some embodiments, the cross-sectional view of FIG. **12C** passes through the intersection position of the grid lines in the grid pattern. The sealing layer **127** can have an interface with the top surface of the dielectric layer **121**. In other words, there is no void between the bottom surface **127b'** of the sealing layer **127** and the top surface of the dielectric structure **125**, and the dielectric structure **125** can have a concave top surface (or concave front-side surface) to be conformal to the bottom surface **127b'** of the sealing layer **127**. Specifically, in FIG. **12C**, when viewed from the cross-sectional view taken along line B-B' (e.g., x-cut) as shown in FIG. **1A**, a thickness **127t3** of the central portion **127c** of the sealing layer **127** can be thicker than thicknesses **127t4** of the peripheral portions **127p**. In some embodiments, a portion of the bottom surface **127b'** of the sealing layer **127** can be exposed from the seam **121s** in the dielectric structure **125**. In some embodiments, the bottom surface **127b'** of the sealing layer **127** can be covered by the dielectric structure **125**.

[0064] The deposition process P7 for forming the sealing layer **127** can be performed in a repetitive cycle, ranging from about 5 to 30 cycles, such as about 5, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, or 30 cycles. Each cycle may include of two sub-steps. The first sub-step can be an epitaxial deposition of the sealing layer **127**, and the second sub-step can be an annealing process, which follows the deposition. The epitaxial deposition may include an epitaxial layer of silicon or a similar material to form the sealing layer **127**. The epitaxial deposition is performed selectively on the exposed sidewalls **102s** of the substrate **102** in the trench **102t**, establishing the initial crystalline structure and thickness of the sealing layer **127**. Following the epitaxial deposition, an annealing process can be applied. The annealing process can include heating the substrate **102** to a high temperature, which in turn helps in relieving stress, repairing defects, improving crystalline quality within the deposited layer, and enhancing the uniformity and density of the sealing layer **127**.

[0065] In some embodiments, repeating the epitaxial deposition and annealing process multiple times can allow for incremental build-up of the sealing layer, enhancing control over its thickness

and uniformity, which in turn helps in gradually improving the crystalline quality of the layer with each cycle, reducing defects and irregularities. Improved epitaxial quality of the sealing layer **127** can ensure a stronger, more uniform layer that can effectively resist etchants and protect underlying layers during subsequent etching processes.

[0066] In some embodiments, the epitaxial deposition of epitaxial layer can be performed at a first temperature, and the annealing can be performed at a second temperature higher than the first temperature. In some embodiments, the first temperature can be in a range from about 600 to 1000° C., such as about 600, 650, 700, 750, 800, 850, 900, 950, or 1000° C. The second temperature can be in a range from about 700 to 1500° C., such as about 700, 800, 900, 1000, 1100, 1200, 1300, 1400, or 1500° C. In some embodiments, the annealing is performing with a mixture gas including H.sub.2. Subsequently, the patterned mask layer **113** can be stripped away (see FIG. **13**). In some embodiments, the patterned mask layer **113** may be removed by dry etching, wet etching, or a combination of dry and wet etching.

[0067] The sealing layer **127**, made of monocrystalline silicon, can be deposited in a controlled manner over the substrate **102**, specifically targeting the area around and above the dielectric layer **121**. The sealing layer **127** can be impermeable and acts as a barrier, preventing the migration of materials into areas where they are not intended. In some embodiments, the seam **121s** within the dielectric layer **121** can be an area where unwanted silicon deposition could occur. If silicon was to deposit in the seam **121s**, it could compromise the electrical isolation and structural integrity of the deep trench isolation (DTI) structure **150** (see FIGS. **25A-25C**), affecting the sensor's performance.

[0068] Reference is made to FIGS. **13** and **14**. A mask layer **133** (see FIG. **13**) is formed over the substrate **102**. Subsequently, a silicon-based material layer **134** (see FIG. **14**) is formed over the mask layer **133**, and a photoresist layer **135** (see FIG. **14**) is formed over the silicon-based material layer **134**. The mask layer **133**, the silicon-based material layer **134**, and the photoresist layer **135** can be used to pattern the sealing layer **127** (see FIG. **14**). In some embodiments, the mask layer **133** may be made of nitride, such as silicon nitride, the silicon-based material layer **134** may be made of SiON, and the photoresist layer **135** may be made of a SiC.sub.xH.sub.yO.sub.z material.

[0069] Reference is made to FIGS. **15A** and **15B**, an etching process **P8** is performed on the substrate **102** and the sealing layer **127** through the mask layer **133**, the silicon-based material layer **134**, and the photoresist layer **135**. Therefore, an upper portion of the sealing layer **127** can be removed, and at least a part of the upper portion of the trench **102t** can reappear and be enlarged to have the lateral dimension **D11**. The lateral dimension **D11** shown in FIG. **14** can be greater than the lateral dimension **D5** shown in FIG. **5**. In some embodiments, after the etching process **P8** is complete, the upper portion of the trench **102t** has a lateral dimension **D11** decreasing as a vertical distance from the front side **102f** of the substrate **102** increases. By way of example and not limitation, the lateral dimension **D11** may be in a range from about 150 to 250 nm, such as 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, or 250 nm. The upper portion of the trench **102t** can have a vertical dimension **D12** that is in a range from about 100 to 200 nm, such as 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 nm. In some embodiments, the lateral dimension **D11** can be greater than the vertical dimension **D12**. Subsequently, the silicon-based material layer **134** and the photoresist layer **135** can be stripped away (see FIG. **15A**). In some embodiments, the silicon-based material layer **134** and the photoresist layer **135** may be removed by dry etching, wet etching, or a combination of dry and wet etching.

[0070] Reference is made to FIG. **16**. A dielectric layer **137** can be deposited over the front side **102f** of the substrate **102** and fill into the trench **102t**. In some embodiments, the dielectric layer **137** can be interchangeably referred to as an oxide layer. By way of example and not limitation, the dielectric layer **137** may be made of a dielectric material such as silicon nitride (SiN), silicon oxide (SiO.sub.2), silicon carbo-nitride (SiCN), silicon oxynitride (SiON), silicon oxy-carbo-nitride (SiOCN), or the like.

[0071] Reference is made to FIG. **17**. An etching process **P9** is performed on the dielectric layer

137 to recess the dielectric layer **137**, such that the dielectric layer **137** can have a top surface lower than the front-side surface of the substrate **102**. In some embodiments, after the etching process **P9** is complete, the dielectric layer **137** can have a top surface in a position between a lower boundary of the node **112** and an upper boundary of the node **112**. In some embodiments, the dielectric layer **137** can have a level back-side surface conformal to the front-side surface of the sealing layer **127**. In some embodiments, the dielectric layer **137** can have a level top surface in parallel with the level bottom surface thereof. The etching process **P9** can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes. In some embodiments, the etching process **P9** may include a first planarization process, such as a chemical mechanical polishing (CMP) process, and a wet etching process, which follows the planarization process.

[0072] Reference is made to FIG. **18**. A semiconductor layer **147** is formed over the substrate **102** and fills into a remainder of the trench **102t** by a deposition process **P10**. In some embodiments, the semiconductor layer **147** can be made of a same material as the sealing layer **127**. By way of example and not limitation, the semiconductor layer **147**, made of polycrystalline silicon (poly-Si), contrasts with the sealing layer **127**, which may be made of monocrystalline silicon. The semiconductor layer **147** of polycrystalline silicon can include of multiple small silicon crystals or grains. The deposition process **P10** can be used to form the semiconductor layer **147** of polycrystalline silicon over the substrate **102**, and thus the semiconductor layer **147** can be interchangeably referred to as a poly silicon layer. In some embodiments, the deposition process **P10** for forming the semiconductor layer **147** is different than the deposition process **P7** for forming the sealing layer **127**. The deposition process **P10** can be performed using techniques like chemical vapor deposition (CVD) process or physical vapor deposition (PVD) process.

[0073] During the deposition process **P10**, silicon atoms are delivered to the substrate surface, where they nucleate and grow, and the silicon atoms form multiple crystalline grains, resulting in a polycrystalline structure. The polycrystalline silicon layer fills the remainder of the trench **102t**, conforming to the shape of the trench **102t**. As the semiconductor layer **147** is deposited, it can adhere to the exposed surfaces within the trench **102t**, including the top of the dielectric layer **137** and the sidewalls **102s** of the trench **102t**. The semiconductor layer **147** of polycrystalline silicon can have ability to conformally coat irregular surfaces (e.g., the inside of the trench **102t**), and its electrical properties, which can be suitable for sensor applications. In some embodiments, the use of semiconductor layer **147** of polycrystalline silicon over sealing layer **127** can be driven by factors like cost-effectiveness, process compatibility, and desired electrical characteristics in the image sensor **100**.

[0074] Reference is made to FIG. **19**. An etching process **P11** is performed on the semiconductor layer **147** to etch back the semiconductor layer **147**. As shown in FIG. **19**, the semiconductor layer **147** is located within the node **112** and can serve as an electrode of the node **112**. The semiconductor layer **147** can be used in the charge-to-voltage conversion process that occurs at the node **112**. Specifically, the electrons collected by the photodiode can be transferred to the node **112**, where they are converted into a voltage signal by the semiconductor layer **147**.

[0075] As shown in FIG. **19**, the dielectric layer **137** is sandwiched between the sealing layer **127** and the semiconductor layer **147**. The dielectric layer **137** can acts as an insulator, preventing electrical crosstalk and leakage between the semiconductor layer **147** and the sealing layer **127**. The dielectric properties of the dielectric layer **137** is for maintaining the integrity of the electrical signals within the node **112** and for ensuring that the charge transfer process can be efficient and free from interference.

[0076] The etching process **P9** can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes. In some embodiments, the etching process **P11** can be a planarization process (e.g., a CMP process, or a grinding process) that is performed to remove portions of the semiconductor layer **147** overlying the mask layer **133** and thin down the semiconductor layer **147**. In some embodiments, the etching process **P11** may also remove portions

of or all the mask layer **133**, such that after the etching process **P11**, the mask layer **133** may have a thickness **T7** thinner than a thickness **T6** of the mask layer **133** shown in FIG. **18**. In some embodiments, the semiconductor layer **147** can have the top surface in a position level with the top surface of the mask layer **133** and higher than the front-side surface of the pad layer **101** and the front-side surface of the substrate **102**.

[0077] In some embodiments, the semiconductor layer **147** may have a vertical dimension (or thickness) **D13** greater than a vertical dimension (or thickness) **D14** of the sealing layer **127**. By way of example and not limitation, the vertical dimension **D13** of the semiconductor layer **147** can be in a range from about 150 to 400 nm, such as about 150, 200, 250, 300, 350, or 400 nm. The vertical dimension **D14** of the sealing layer **127** can be in a range from about 20 to 100 nm, such as 20, 30, 40, 50, 60, 70, 80, 90, or 100 nm. In some embodiments, a combination of the thicknesses of the dielectric layer **137** and the semiconductor layer **147** can have a vertical dimension **D15**. The vertical dimension **D15** can be greater than the vertical dimension **D14** of the sealing layer **127**. By way of example and not limitation, the vertical dimension **D15** can be in a range from about 100 to 400 nm, such as about 100, 150, 200, 250, 300, 350, or 400 nm. As shown in FIG. **19**, the sealing layer **127** can be formed with a footprint that is disposed within a footprint of the dielectric layer **137** and/or a footprint of the semiconductor layer **147**. Subsequently, the mask layer **133** can be stripped away (see FIG. **20**). In some embodiments, the mask layer **133** may be removed by dry etching, wet etching, or a combination of dry and wet etching.

[0078] Reference is made to FIG. **21**. A transfer gate **114** can be formed over/on the front side **102f** of the substrate **102**. The plurality of transfer gates are formed overlying, at least partially, the plurality of pixel regions **103** (see FIG. **1A**), respectively. In some embodiments, each of the plurality of transfer gates **104** can be formed between two portions of the doped well **124**. Each of the plurality of transfer gates **114** are formed with a gate dielectric structure **116** and a gate electrode structure **118**. A spacer **117** can be formed on opposite sidewalls of the gate electrode structure **118** or laterally surround the gate electrode structure **118**.

[0079] In some embodiments, a process for forming the plurality of transfer gates **114** can include forming a first patterned masking layer (not shown) (e.g., negative/positive photoresist, a hardmask, etc.) over the front side **102f** of the substrate **102**. With the first patterned masking layer in place over the front side **102f** of the substrate **102**, an etching process is then performed on the substrate **102**. The etching process removes unmasked portions of the substrate **102**, thereby forming a plurality of vertical gate trenches in the substrate **102**. The etching process may be or comprise, for example, a wet etching process, a dry etching process, a RIE process, some other etching process, or a combination of the foregoing. Subsequently, in some embodiments, the patterned masking layer is then stripped away.

[0080] Thereafter, a gate dielectric layer (not shown) can be formed over/on the front side **102f** of the substrate **102** and lining the plurality of vertical gate trenches. In some embodiments, the gate dielectric layer can be or include, for example, an oxide (e.g., silicon dioxide (SiO_2)), a high-k dielectric material (e.g., hafnium oxide (HfO_2), tantalum oxide (TaO_2), hafnium silicon oxide (HfSiO_2), hafnium tantalum oxide (HfTaO_2), aluminum oxide (AlO_2), zirconium oxide (ZrO_2), some other dielectric material with a dielectric constant greater than about 3.9), some other dielectric material, or a combination of the foregoing. A gate electrode layer (not shown) can be then formed on the gate dielectric layer and in the plurality of vertical gate trenches. In some embodiments, the gate electrode layer can be or include, for example, polysilicon, a metal (e.g., aluminum (Al), copper (Cu), titanium (Ti), tantalum (Ta), tungsten (W), molybdenum (Mo), cobalt (Co), or the like), some other conductive material, or a combination of the foregoing.

[0081] Thereafter, a second patterned masking layer (not shown) (e.g., negative/positive photoresist, a hardmask, etc.) is formed over the gate electrode layer. With the second patterned masking layer in place, an etching process is performed on the gate electrode layer and the gate dielectric layer. The etching process removes unmasked portions of the gate electrode layer,

thereby forming the gate electrode structures (see, e.g., the gate electrode structure **118** described herein). The etching process also removes unmasked portions of the gate dielectric layer, thereby forming the gate dielectric structures (see, e.g., the gate dielectric structure **116** described herein). In some embodiments, the etching process may be or comprise, for example, a wet etching process, a dry etching process, a RIE process, some other etching process, or a combination of the foregoing. Subsequently, in some embodiments, the second patterned masking layer is stripped away. In some embodiments, the spacer **117** may be or include, for example, a high-k dielectric material (e.g., HfO, TaO, HfSiO, HfTaO, AlO, ZrO, etc.), an oxide (e.g., SiO₂), a nitride (e.g., SiN), an oxy-nitride (e.g., SiON), a carbide (e.g., silicon carbide (SiC)), some other dielectric material, or a combination of the foregoing.

[0082] Reference is made to FIG. **22**. An ILD structure **140** is formed over the front side **102f** of the substrate **102** and over the plurality of transfer gates **114**. An interconnect structure **142** is formed in the ILD structure **140** and over the front side **102f** of the substrate **102**. In some embodiments, the interconnect structure **142** can include a plurality of conductive contacts **142a**, a plurality of conductive vias **142b**, and a plurality of conductive wires **142c**.

[0083] In some embodiments, a process for forming the ILD structure **140** and the interconnect structure **142** comprises forming a first ILD layer over the front side **102f** of the substrate **102**. Thereafter, contact openings are formed in the first ILD layer. A conductive material (e.g., tungsten (W), copper (Cu), aluminum (Al)) is then formed on the first ILD layer and in the contact openings. Thereafter, a planarization process (e.g., chemical-mechanical planarization (CMP)) is performed on the conductive material to form the plurality of conductive contacts **142a** in the first ILD layer. A second ILD layer is then formed over the first ILD layer and the plurality of conductive contacts **142a**. A plurality of trenches are then formed in the second ILD layer. A conductive material (e.g., copper (Cu)) is formed on the second ILD layer and in the trenches. Thereafter, a planarization process (e.g., CMP) is performed into the conductive material to form a first group of conductive wires of the plurality of conductive wires **142c** (e.g., the conductive wires of a first layer of conductive wires (e.g., metal layer **1**)).

[0084] Thereafter, the plurality of conductive vias **142b** and the remaining conductive wires of the plurality of conductive wires **142c** of the interconnect structure **142** may be formed by repeating a damascene process (e.g., a single damascene process or a dual damascene process) until a predefined number of conductive vias and conductive wires are formed in the ILD structure **140**. The damascene process may be performed by depositing a subsequent ILD layer over the second ILD layer and the first group of conductive wires of the plurality of conductive wires **142c**, etching the subsequent ILD layer to form one or more via holes and/or one or more trenches in the subsequent ILD layer, and filling the one or more via holes and/or the one or more trenches with a conductive material (e.g., copper (Cu)). Thereafter, a planarization process (e.g., CMP) is performed on the conductive material, thereby forming a second group of conductive wires of the plurality of conductive wires **142c** (e.g., the conductive wires of a second layer of conductive wires (e.g., metal layer **2**)) and/or the conductive vias of the plurality of conductive vias **142b** that extend vertically between the first group and the second group of conductive wires. This damascene process is repeated until each of the plurality of conductive vias **142b** and each of the plurality of conductive wires **142c** of the interconnect structure **142** are formed in the ILD structure **140**. The ILD layers may be formed by, for example, CVD, PVD, ALD, some other deposition process, or a combination of the foregoing. The conductive material(s) (e.g., tungsten (W), copper (Cu), etc.) may be formed using a deposition process (e.g., CVD, PVD, sputtering, etc.) and/or a plating process (e.g., electrochemical plating, electroless plating, etc.).

[0085] Reference is made to FIG. **23**. The structure of FIG. **22** is “flipped” upside down. Subsequently, the substrate **102** can be thinned down through at least one removal process, for example, CMP, HNA, and/or TMAH etching from the back side **102b** of the substrate **102**, and thus the dielectric layer **121** can be exposed from the back side **102b** of the substrate **102**. In some

embodiments, after the substrate **102** is thinned down, the substrate may have a thickness in a range from about 1000 to 5000 nm, such as about 1000, 1500, 2000, 2670, 2500, 3000, 3500, 4000, 4500, or 5000 nm. Subsequently, the dielectric layer **121** is removed from the back side **102b** of the substrate **102** through a removal process **P12**. During the removal process **P12**, the sealing layer **127** may be used as etch stop layers when the dielectric layer **121** is etched. In some embodiments, the removal process **P12** may include, for example, a wet etching process, a dry etching process, a reactive ion etching (RIE) process, some other etching process, or a combination of the foregoing. The removal process **P12** can selectively etch the dielectric layer **121** at a faster rate than the substrate **102** and the sealing layer **127**. By way of example but not limiting the present disclosure, the removal process **P12** may be performed using dilute hydrofluoric (dHF) acid.

[0086] Reference is made to FIG. **24**. A dielectric liner **152** is formed lining surfaces of the trench **102t** (e.g., sidewalls of the trench **102t**, lower surfaces of the trench **102t**, etc.) from the back side **102b** of the substrate **102**. In some embodiments, the dielectric liner **152** is formed contacting (e.g., directly contacting) the first doped region **104**, the second doped region **106**, the node **112**, and the doped well **124**. In some embodiments, the dielectric liner **152** can be omitted. In some embodiments, a process for forming the dielectric liner **152** can include depositing a dielectric liner layer (not shown) over the back side **102b** of the substrate **102** and along the surfaces of the trench **102t**. The dielectric liner layer may be or include, for example, a high-k dielectric material (e.g., HfO, TaO, HfSiO, HfTaO, AlO, ZrO, etc.), an oxide (e.g., SiO.sub.2), a nitride (e.g., SiN), an oxynitride (e.g., SiON), a carbide (e.g., silicon carbide (SiC)), some other dielectric material, or a combination of the foregoing. The dielectric liner layer may be deposited by, for example, CVD, PVD, ALD, some other deposition process, or a combination of the foregoing. Thereafter, an upper portion of the dielectric liner layer over the back side **102b** of the substrate **102** can be removed, thereby leaving remaining portions in place as the dielectric liner **152**. In some embodiments, the upper portion of the dielectric liner layer may be removed by, for example, a planarization process (e.g., chemical-mechanical planarization (CMP)), an etching process (e.g., wet etching, dry etching, etc.), some other removal process, or the like.

[0087] Reference is made to FIGS. **25A-25C**. A dielectric filler **154** is deposited over the back side **102b** of the substrate **102** and fills into the trench **102t**. The dielectric filler **154** can be used to provide electrical insulation and/or structural integrity to the image sensor **100**. In some embodiments, forming the dielectric filler **154** over the dielectric liner **152** can complete formation of a deep trench isolation (DTI) structure **150** in the trench **102t**. In other words, the dielectric liner **152** and the dielectric filler **154** in the trench **102t** can be collectively referred to the deep trench isolation structure **150** that may inherit the profile of the trench **102t**. In some embodiments, the dielectric filler **154** over the back side **102b** of the substrate **102** may have a thickness in a range from about 50 to 300 nm, such as about 50, 100, 143, 150, 200, 250, or 300 nm.

[0088] In the cross-sectional view, the deep trench isolation structure **150** can be characterized by the positioning of the node **112** directly above the deep trench isolation structure **150**. The deep trench isolation structure **150** can be a deeply etched trench in the substrate **102**, which is filled with dielectric materials to isolate the components of the sensor from each other, thereby reducing electronic noise and crosstalk between pixels. When viewed from the top, the deep trench isolation structure **150** can reveal that the node **112** is located at the intersection (or cross-road) of the isolation grid **156** (see FIG. **26**).

[0089] In some embodiments, the deep trench isolation structure **150** can be interchangeably referred to a floating diffusion trench isolation (FDTI), and thus the image sensor **100** can be interchangeably referred to a FDTI-CMOS image sensor (FDTI-CIS). The deep trench isolation structure **150** is formed with a footprint that is disposed within the footprint of the doped well **124**, which in turn reduces the size of the pixel region **103**. As shown in FIGS. **25B** and **25C**, the sealing layer **127** is positioned between the node **112** and the deep trench isolation structure **150**. The sealing layer **127** can act as a barrier or an etch stop layer during the manufacturing process to

protect the underlying structures during subsequent etching and processing steps (see FIG. 23). In some embodiments, a planarization process (e.g., CMP) can be performed on the dielectric filler **154** to planarize a back-side surface of the dielectric filler **154**. In further embodiments, the planarization process (and/or some other removal process) can be performed on the dielectric filler **154** to remove a portion of the dielectric filler **154** over the back side **102b** of the substrate **102**. [0090] As shown in FIGS. 25A-25C, the deep trench isolation structure **150** can vertical sidewalls, which indicate the sidewalls of the deep trench isolation structure **150** formed in the substrate **102** can be perpendicular to the front-side surface of the substrate **102** and/or the back-side surface of the substrate **102**. This vertical alignment can be used to create well-defined, isolated regions for each pixel in the sensor array. As shown in FIG. 25B, when viewed from the cross-sectional view taken along line A-A' (e.g., xy-cut) as shown in FIG. 1A, the deep trench isolation structure **150** can have a flat front-side surface perpendicular to the sidewalls thereof, and thus the deep trench isolation structure **150** can have a square front-side profile. This square profile shown in FIG. 25B is in direct contact with and conformal to the sealing layer **127**. As shown in FIG. 25C, when viewed from the cross-sectional view taken along line B-B' (e.g., x-cut) as shown in FIG. 1A, the deep trench isolation structure **150** can have a concave front-side surface, and thus a front-side of the deep trench isolation structure **150** can have a concave front-side profile. This concave profile shown in FIG. 25C is in direct contact with and conformal to the sealing layer **127**. In some embodiments, the sidewall of the deep trench isolation structure **150** is coterminous to the sidewall of the sealing layer **127**.

[0091] The dielectric filler **154** can be formed to have a seam **154s** therein within the trench **102t**. In some embodiments, the seam **154s** can be interchangeably referred to as an air gap. In some embodiments, the seam **154s** may have a first distance to the front side **102f** of the substrate **102**, and the first distance can be in a range from about 60 to 100 nm, such as about 60, 65, 70, 75, 80, 85, 90, 95, or 100 nm. In some embodiments, the seam **154s** may have a second distance to the back side **102b** of the substrate **102**, and the second distance can be in a range from about 20 to 50 nm, such as about 20, 25, 28, 30, 35, 40, 45, or 50 nm.

[0092] By way of example and not limitation, the dielectric filler **154** may be made of a dielectric material such as silicon nitride (SiN), silicon oxide (SiO.sub.2), silicon carbo-nitride (SiCN), silicon oxynitride (SiON), silicon oxy-carbo-nitride (SiOCN), or the like. The dielectric layer **123** may be made of a dielectric material such as silicon nitride (SiN), silicon oxide (SiO.sub.2), silicon carbo-nitride (SiCN), silicon oxynitride (SiON), silicon oxy-carbo-nitride (SiOCN), or the like. In some embodiments, the trench **102t** can be filled only with dielectric filler **154** but not with the dielectric liner **152**. In some embodiments, the dielectric filler **154** can be interchangeably referred to a dielectric layer or an isolation layer. In some embodiments, the dielectric filler **154** can be deposited using a plasma enhanced atomic layer deposition (PEALD) process. The plasma enhanced atomic layer deposition process can use plasma to enhance chemical reactions at the surface to form the dielectric filler **154**. By using the plasma enhanced atomic layer for deposition, the dielectric filler **154** can adhere well to the underlying structures, maintains uniform thickness across the substrate **102**. The use of plasma in the ALD process can also enhance the quality of the dielectric layer by improving its density and uniformity.

[0093] The deep trench isolation structure **150** extends partially through the substrate **102** (e.g., not fully through the substrate **102**) from the back side **102b** of the substrate **102**, thereby extending vertically through doped well **124**, the second doped region **106**, and the first doped region **104**. In such embodiments, the deep trench isolation structure **150** can expose portions of the first doped region **104** and portions of the second doped region **106**. In some embodiments, the deep trench isolation structure **150** may have angled sidewalls. In other embodiments, the sidewalls of the deep trench isolation structure **150** may be substantially straight (e.g., vertical).

[0094] The deep trench isolation structure **150** is formed extending laterally through the substrate **102**, such that the deep trench isolation structure **150** laterally surrounds the plurality of pixel

regions **103**. The deep trench isolation structure **150** is formed with a footprint that is disposed within the footprint of the doped well **124**. In some embodiments, the second doped region **106** continuously extends laterally between opposite inner sidewalls of the deep trench isolation structure **150**. In some embodiments, the second doped region **106** continuously extends laterally between the opposite inner sides of the doped well **124** and continuously extends laterally between the opposite inner sidewalls of the deep trench isolation structure **150**. In some embodiments, the first doped region **104** continuously extends laterally between the opposite inner sidewalls of the deep trench isolation structure **150**. In some embodiments, a layout of the trench **102t** has a grid-like shape. As such, the footprint of the deep trench isolation structure **150** has the grid-like shape. In such embodiments, the grid-like shaped footprint of the deep trench isolation structure **150** is disposed within a grid-like shaped footprint of the doped well **124**.

[0095] Reference is made to FIG. **26**. An isolation grid **156** is formed over a back-side surface of the dielectric filler **154** over the substrate **102**. In other embodiments, the isolation grid **156** may be formed along the back side **102b** of the substrate **102**. In further embodiments, the isolation grid **156** can be formed overlying the deep trench isolation structure **150**. In some embodiments, a process for forming the isolation grid **156** may include forming a patterned masking layer (not shown), which has a trench disposed therein, over the back-side surface of the dielectric filler **154**. Thereafter, an isolation material is deposited on the patterned masking layer and in the trench. The isolation material may be or include, for example, a metal (e.g., tungsten (W), aluminum (Al), cobalt (Co), copper (Cu), silver (Ag), gold (Au), some other metal, or a combination of the foregoing), an oxide (e.g., SiO₂), a nitride (e.g., SiN), a carbide (e.g., SiC), a high-k dielectric material (e.g., HfO, TaO, etc.), a low-k dielectric material, some other isolation material, or a combination of the foregoing. Thereafter, a planarization process is performed (e.g., CMP, etch back process, etc.) on the isolation material to remove an upper portion of the isolation material, thereby leaving lower portions of the isolation material in the trench as the isolation grid **156**. Subsequently, in some embodiments, the patterned masking layer can be stripped away.

[0096] A dielectric layer **158** is formed over the over the back-side surface of the dielectric filler **154** and over the isolation grid **156**. In other embodiments, the dielectric layer **158** may be formed along the back side **102b** of the substrate **102** and over the isolation grid **156**. The dielectric layer **158** may be deposited to form by, for example, CVD, PVD, ALD, sputtering, some other deposition process, or a combination of the foregoing.

[0097] Reference is made to FIG. **27**. A plurality of electromagnetic radiation (EMR) filters **160** are formed on/over the dielectric layer **158** and within the isolation grid **156**. In some embodiments, a process for forming the plurality of EMR filters **160** can include depositing (e.g., via CVD, PVD, ALD, sputtering, a spin-on process, etc.) one or more light filtering materials onto the dielectric layer **158** and within the isolation grid **156**. The one or more light filtering materials are materials that allow for the transmission of radiation (e.g., light) having a specific wavelength range, while blocking light of wavelengths outside of the specified range. Subsequently, in some embodiments, a planarization process (e.g., CMP) may be performed on the plurality of EMR filters **160** to planarize the upper surface of the plurality of EMR filters.

[0098] Reference is made to FIGS. **28A** and **28B**. A plurality of micro-lenses **162** are formed on/over the plurality of EMR filters **160**. In some embodiments, the plurality of micro-lenses **162** may be formed by depositing a micro-lens material on the plurality of EMR filters **160** (e.g., via CVD, PVD, ALD, sputtering, a spin-on process, etc.). A micro-lens template (not shown) having a curved upper surface is patterned above the micro-lens material. In some embodiments, the micro-lens template may comprise a photoresist material exposed using a distributing exposing light dose (e.g., for a negative photoresist more light is exposed at a bottom of the curvature and less light is exposed at a top of the curvature), developed, and baked to form a rounding shape. The plurality of micro-lenses are then formed by selectively etching the micro-lens material according to the micro-lens template. In some embodiments, after the plurality of micro-lenses **162** are formed, formation

of a plurality of pixel sensor units **107** (see FIG. **1D**) is complete. In further embodiments, after the plurality of micro-lenses **162** are formed, formation of the image sensor **100** can be complete. In some embodiments, the image sensor **100** can be interchangeably referred to as a back-side illumination configuration.

[0099] Therefore, based on the above discussions, it can be seen that the present disclosure offers advantages. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. The present disclosure in various embodiments provides a FDTI-CIS using selective silicon epitaxial growth (SEG) for the sealing layer to prevent polysilicon deposition in the oxide seam of FDTI structure. The SEG can be utilized for control of the sealing layer's profile over the FDTI structure, ensuring accurate layer formation. The sealing layer can be in a crystalline state to have a higher etching resistance to prevent the etchant from damaging the dielectric layer over the sealing layer. In addition, before forming the sealing layer over the oxide of the FDTI structure, a dry etch process (e.g., capacitively coupled plasma) can create a thin oxide layer on the FDTI sidewall. Subsequently, a weak pre-clean process (e.g., SiCoNi) followed by cyclic deposition and annealing can be incorporated to scale down the thin oxide layer. Therefore, the sealing layer can form a flat Si-EPI merged-plane over the thin oxide layer, which in turn allowing for yielding a high-quality sealing layer, and thus the performance of the image sensor can be enhanced.

[0100] In some embodiments, a method includes forming a doped region in a semiconductor substrate, wherein the doped region is of a first conductivity type; forming a doped well vertically extending from a front side of the semiconductor substrate into the doped region, wherein the doped well is of a second conductivity type opposite to the first conductivity type; forming a transfer gate over the front side of the semiconductor substrate and laterally between a first portion of the doped well and a second portion of the doped well; forming a trench vertically extending from the front side of the semiconductor substrate through the doped well toward a back side of the semiconductor substrate; depositing a dielectric material over the front side of the semiconductor substrate and in the trench; recessing the dielectric material, such that the recessed dielectric material has a top surface lower than the front side of the semiconductor substrate; performing a selective epitaxial growth process to form a sealing layer in the trench and over the recessed dielectric material from the front side of the semiconductor substrate; forming an electrode in the trench and over the sealing layer from the front side of the semiconductor substrate. In some embodiments, the step of recessing the dielectric material comprises: performing a planarization process to remove the dielectric material located outside the trench; after performing the planarization process, performing a dry etching process to recess the dielectric material. In some embodiments, the dry etching process comprises an etchant including C.sub.4F.sub.8. In some embodiments, the dry etching process is performed such that the recessed dielectric material has a sharp protrusion on a top thereof. In some embodiments, the method further includes after performing the dry etching process, performing a pre-clean process on the recessed dielectric material to scale down the sharp protrusion. In some embodiments, the pre-clean process comprises an etchant including HF and NH.sub.3. In some embodiments, the selective epitaxial growth process comprises a repetitive cycle, and the each cycle comprises an epitaxial deposition step and an annealing step following the epitaxial deposition step. In some embodiments, the method further includes forming a floating diffusion node in the first portion of the doped well, wherein the electrode is formed in the floating diffusion node. In some embodiments, the method further includes after forming the electrode, removing the dielectric material in the trench from the back side of the semiconductor substrate to expose the sealing layer; after removing the dielectric material, forming a high-k dielectric liner lining a sidewall of the trench from the back side of the semiconductor substrate; after forming the high-k dielectric liner, forming a dielectric filler in the trench from the back side of the semiconductor substrate to form a deep trench isolation structure.

In some embodiments, the method further includes forming a plurality of micro-lenses over the back side of the semiconductor substrate.

[0101] In some embodiments, a method includes forming a doped well vertically extending from a front side of a substrate into the substrate and surrounding a pixel region of an image sensor; forming a trench vertically extending from the front side of the substrate through the doped well toward a back side of the substrate; forming a dielectric structure in the trench from the front side of the substrate; performing a selective epitaxial growth process to form an epitaxial silicon layer in the trench and over a front side of the dielectric structure, wherein the selective epitaxial growth process comprises a repetitive cycle, and each cycle comprises an epitaxial deposition step and an annealing step following the epitaxial deposition step; forming a dielectric layer in the trench and over a front side of the epitaxial silicon layer; forming a polysilicon layer in the trench and over a front side of the dielectric layer; replacing the dielectric structure with a deep trench isolation structure from the back side of the substrate. In some embodiments, the epitaxial deposition step in the selective epitaxial growth process is performed at a temperature in a range from about 600 to 900° C. In some embodiments, forming the dielectric structure includes depositing a dielectric material over the front side of the semiconductor substrate and in the trench; recessing the dielectric material, such that the recessed dielectric material has a top surface lower than the front side of the semiconductor substrate. In some embodiments, from a top view, the deep trench isolation structure has a grid pattern that has a grid line, and from a cross-sectional view taken along a direction intersecting with a lengthwise direction of the grid line, the epitaxial silicon layer has a flat back-side surface. In some embodiments, from a top view, the deep trench isolation structure has a grid pattern that has a grid line, and from a cross-sectional view taken along a lengthwise direction of the grid line, the epitaxial silicon layer has a convex back-side surface. In some embodiments, the step of replacing the dielectric structure with the deep trench isolation structure comprises: thinning down the substrate to expose the dielectric structure from the back side of the substrate; removing the dielectric structure in the trench from the back side of the substrate; forming the deep trench isolation structure in the trench.

[0102] In some embodiments, an image sensor includes a first conductivity type doped region, a second conductivity type doped well, a deep trench isolation structure, an epitaxial sealing layer, and a polysilicon layer. The first conductivity type doped region is in a pixel region of a substrate. The second conductivity type doped well vertically extends from a front side of the substrate into the first conductivity type doped region, wherein from a top view, the second conductivity type doped well laterally surrounds the pixel region. The deep trench isolation structure is in the substrate and laterally surrounds the pixel region, wherein a footprint of the deep trench isolation structure is within a footprint of the second conductivity type doped well. The epitaxial sealing layer is over a front side of the deep trench isolation structure and in the second conductivity type doped well. The polysilicon layer is over a front side of the epitaxial sealing layer and embedded in the second conductivity type doped well. In some embodiments, the epitaxial sealing layer has a flat back side surface. In some embodiments, the epitaxial sealing layer has a thickness in a range from about **20** to **100** nm. In some embodiments, the deep trench isolation structure has a flat front side surface. In some embodiments, the deep trench isolation structure has a vertical sidewall wall.

[0103] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming a doped region in a semiconductor substrate, wherein the doped region is of a first conductivity type; forming a doped well vertically extending from a front side of the semiconductor substrate into the doped region, wherein the doped well is of a second conductivity type opposite to the first conductivity type; forming a transfer gate over the front side of the semiconductor substrate and laterally between a first portion of the doped well and a second portion of the doped well; forming a trench vertically extending from the front side of the semiconductor substrate through the doped well toward a back side of the semiconductor substrate; depositing an dielectric material over the front side of the semiconductor substrate and in the trench; recessing the dielectric material, such that the recessed dielectric material has a top surface lower than the front side of the semiconductor substrate; performing a selective epitaxial growth process to form a sealing layer in the trench and over the recessed dielectric material from the front side of the semiconductor substrate; and forming an electrode in the trench and over the sealing layer from the front side of the semiconductor substrate.
2. The method of claim 1, wherein the step of recessing the dielectric material comprises: performing a planarization process to remove the dielectric material located outside the trench; and after performing the planarization process, performing a dry etching process to recess the dielectric material.
3. The method of claim 2, wherein the dry etching process comprises an etchant including C.sub.4F.sub.8.
4. The method of claim 2, wherein the dry etching process is performed such that the recessed dielectric material has a sharp protrusion on a top thereof.
5. The method of claim 2, further comprising: after performing the dry etching process, performing a pre-clean process on the recessed dielectric material to scale down the sharp protrusion.
6. The method of claim 5, wherein the pre-clean process comprises an etchant including HF and NH.sub.3.
7. The method of claim 1, wherein the selective epitaxial growth process comprises a repetitive cycle, and the each cycle comprises an epitaxial deposition step and an annealing step following the epitaxial deposition step.
8. The method of claim 1, further comprising: forming a floating diffusion node in the first portion of the doped well, wherein the electrode is formed in the floating diffusion node.
9. The method of claim 1, further comprising: after forming the electrode, removing the dielectric material in the trench from the back side of the semiconductor substrate to expose the sealing layer; after removing the dielectric material, forming a high-k dielectric liner lining a sidewall of the trench from the back side of the semiconductor substrate; and after forming the high-k dielectric liner, forming a dielectric filler in the trench from the back side of the semiconductor substrate to form a deep trench isolation structure.
10. The method of claim 1, further comprising: forming a plurality of micro-lenses over the back side of the semiconductor substrate.
11. A method, comprising: forming a doped well vertically extending from a front side of a substrate into the substrate and surrounding a pixel region of an image sensor; forming a trench vertically extending from the front side of the substrate through the doped well toward a back side of the substrate; forming a dielectric structure in the trench from the front side of the substrate; performing a selective epitaxial growth process to form a epitaxial silicon layer in the trench and over a front side of the dielectric structure, wherein the selective epitaxial growth process comprises a repetitive cycle, and the each cycle comprises an epitaxial deposition step and an annealing step following the epitaxial deposition step; forming a dielectric layer in the trench and over a front side of the epitaxial silicon layer; forming a polysilicon layer in the trench and over a

front side of the dielectric layer; and replacing the dielectric structure with a deep trench isolation structure from the back side of the substrate.

12. The method of claim 11, wherein the epitaxial deposition step in the selective epitaxial growth process is performed at a temperature in a range from about **600** to **900° C**.

13. The method of claim 11, wherein forming the dielectric structure comprises: depositing an dielectric material over the front side of the semiconductor substrate and in the trench; and recessing the dielectric material, such that the recessed dielectric material has a top surface lower than the front side of the semiconductor substrate.

14. The method of claim 11, wherein from a top view, the deep trench isolation structure has a grid pattern that has a grid line, and from a cross-sectional view taken along a direction intersecting with a lengthwise direction of the grid line, the epitaxial silicon layer has a flat back-side surface.

15. The method of claim 11, wherein from a top view, the deep trench isolation structure has a grid pattern that has a grid line, and from a cross-sectional view taken along a lengthwise direction of the grid line, the epitaxial silicon layer has a convex back-side surface.

16. The method of claim 11, wherein the step of replacing the dielectric structure with the deep trench isolation structure comprises: thinning down the substrate to expose the dielectric structure from the back side of the substrate; removing the dielectric structure in the trench from the back side of the substrate; and forming the deep trench isolation structure in the trench.

17. An image sensor, comprising: a first conductivity type doped region in a pixel region of a substrate; a second conductivity type doped well vertically extending from a front side of the substrate into the first conductivity type doped region, wherein from a top view, the second conductivity type doped well laterally surrounds the pixel region; a deep trench isolation structure in the substrate and laterally surrounding the pixel region, wherein a footprint of the deep trench isolation structure is within a footprint of the second conductivity type doped well; an epitaxial sealing layer over a front side of the deep trench isolation structure and in the second conductivity type doped well; and a polysilicon layer over a front side of the epitaxial sealing layer and embedded in the second conductivity type doped well.

18. The image sensor of claim 17, wherein the epitaxial sealing layer has a flat back side surface.

19. The image sensor of claim 17, wherein the deep trench isolation structure has a flat front side surface.

20. The image sensor of claim 17, wherein the deep trench isolation structure has a vertical sidewall wall.
